Employee Performance and Salary Data Analysis System

PROJECT REPORT

Submitted by:

23BCA10677 Shashi Kant

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Shilpa Mahajan

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Abstract

The Employee Performance and Salary Data Analysis System is a Microsoft Excel-based application designed to efficiently manage and analyze employee data within a single workbook. It allows users to store employee details, monitor performance, and evaluate salary information in an organized manner. The system uses interconnected worksheets and advanced Excel functions like COUNTIF, SUMIF, and AVERAGEIF to ensure automatic updates and accurate analysis.

It also incorporates data visualization tools such as charts and conditional formatting to present insights clearly. The project demonstrates the practical use of Excel for HR analytics in an offline environment and provides a foundation for future enhancements.

Key Points:

- Centralized employee data management
- Automated calculations using Excel formulas
- Performance and KPI tracking
- Salary and department-wise analysis
- Visual representation using charts and formatting
- Offline and easy-to-use system
- Scalable for future integration (Power BI, web, database)

CHAPTER 1: INTRODUCTION

Introduction:

- Employee performance and salary management is crucial for organizational decision-making.
- Manual or scattered data leads to **errors, inconsistency, and data loss**.
- Mid-sized organizations often lack **advanced ERP systems**, creating a need for a simple solution.
- Microsoft Excel provides a **cost-effective and powerful tool** for data management and analysis.
- This system integrates all employee data into a **single, structured Excel workbook**.
- Uses **multiple interconnected sheets** for automatic updates and analysis.
- Includes **formulas, charts, and conditional formatting** for better insights.

Motivation:

- HR data is often **unorganized and difficult to analyze**.
- Manual report generation is **time-consuming**.
- Need for an **automated, accurate, and easy-to-use system**.
- Demonstrates how Excel can be used as a **decision-support tool**.
- Shows integration of Excel with **Python (openpyxl)** for automation.

Project Objectives:

- Store and manage **complete employee data** in one system.
- Perform **automatic calculations** (salary, bonus, performance).
- Generate **department-wise and performance reports**.
- Design a **user-friendly and professional interface**.
- Use **conditional formatting** for performance visualization.
- Include **charts for graphical analysis**.
- Apply advanced Excel functions (**SUMIF, COUNTIF, etc.**).
- Provide a **scalable system** for future upgrades (BI tools, web apps).

CHAPTER 2: Problem Statement / Objectives

Problem Statement

Most organizations — particularly those at the small-to-medium scale — find it extremely difficult to maintain a consolidated, accurate, and analytically useful record of employee performance and salary data. Traditional approaches rely on individual spreadsheets maintained by different managers, paper-based performance review forms, or expensive ERP systems that require significant IT infrastructure and specialized training. Each of these approaches comes with major drawbacks.

Manual spreadsheets created ad hoc by individual managers are rarely consistent in their structure, lack cross-referencing capabilities, and must be manually compiled into summary reports. Paper-based systems provide no analytical capability whatsoever and are prone to physical loss or damage. Enterprise systems, while powerful, are often too expensive and complex for organizations that simply need a reliable way to track 50 to 500 employees.

The consequence of this data fragmentation is that critical HR decisions — such as identifying employees ready for promotion, detecting departments with below-average performance, or comparing compensation across job levels — require hours of manual data gathering and calculation. This leads to delayed decision-making, potential salary inequities, missed promotion opportunities for deserving employees, and an overall lack of data-driven HR management.

Therefore, there is a clear and pressing need for a simple, formula-driven, offline application that allows HR teams and managers to record, analyze, and visualize the complete employee performance and compensation picture from a single, well-structured workbook. The solution should be accessible without specialized software, maintainable without programming knowledge, and powerful enough to support real business decisions.

Objectives

1. To create a fully offline, Excel-based employee performance and salary tracking system that requires no internet connectivity, external software licenses, or database infrastructure to operate.
2. To allow HR users to add, view, and manage employee records dynamically across eight departments and six job levels, with all derived fields — bonus, total compensation, promotion eligibility, and salary band — calculated automatically via formulas.
3. To maintain complete data consistency and integrity through cross-sheet formula-driven calculations, ensuring that any change made in the Employee Database sheet is immediately and accurately reflected across all four downstream analytical sheets.
4. To display department-wise and level-wise categorized summaries covering total headcount, average KPI scores, total payroll costs, average attendance rates, total bonus expenditure, and promotion-eligible employee counts.
5. To provide visual analytical tools including embedded bar charts, 3-color heat map conditional formatting, and the Executive Dashboard KPI card layout for instant data comprehension without the need for manual filtering or pivot tables.
6. To support salary analysis by computing minimum, maximum, average, and standard deviation of base salaries and total compensation for each employment level and salary band, enabling evidence-based compensation benchmarking and pay equity assessment.
7. To ensure that the workbook is simple to understand for non-technical HR users, portable across Windows, macOS, and Linux platforms, and scalable to accommodate larger employee datasets without restructuring the formula architecture.

CHAPTER 3: System Design / Approach

System Overview:

- The system follows a **modular, multi-sheet Excel architecture**.
- Data is divided across **five interconnected worksheets**.
- Ensures **clear structure, accuracy, and easy maintenance**.
- Uses **unidirectional data flow** (Input → Processing → Output).
- Eliminates errors by avoiding circular references.

System Architecture Layers:

1. Input Layer (Employee Database):

- Acts as the **main data source**.
- Stores employee details like **ID, name, department, salary, performance, etc.**
- Uses formulas for **bonus, total compensation, promotion eligibility, and salary band**.
- Includes **conditional formatting and structured design** for better readability.

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2. Processing Layer (Department Summary & Performance Analytics):

- Performs **data aggregation and analysis**.
- Uses functions like **COUNTIF, SUMIF, AVERAGEIF, SUMPRODUCT**.
- Generates **department-wise reports** (salary, KPI, attendance, etc.).
- Includes **charts for visualization** (bar charts, column charts).
- Analyzes **performance levels and KPI distribution**.

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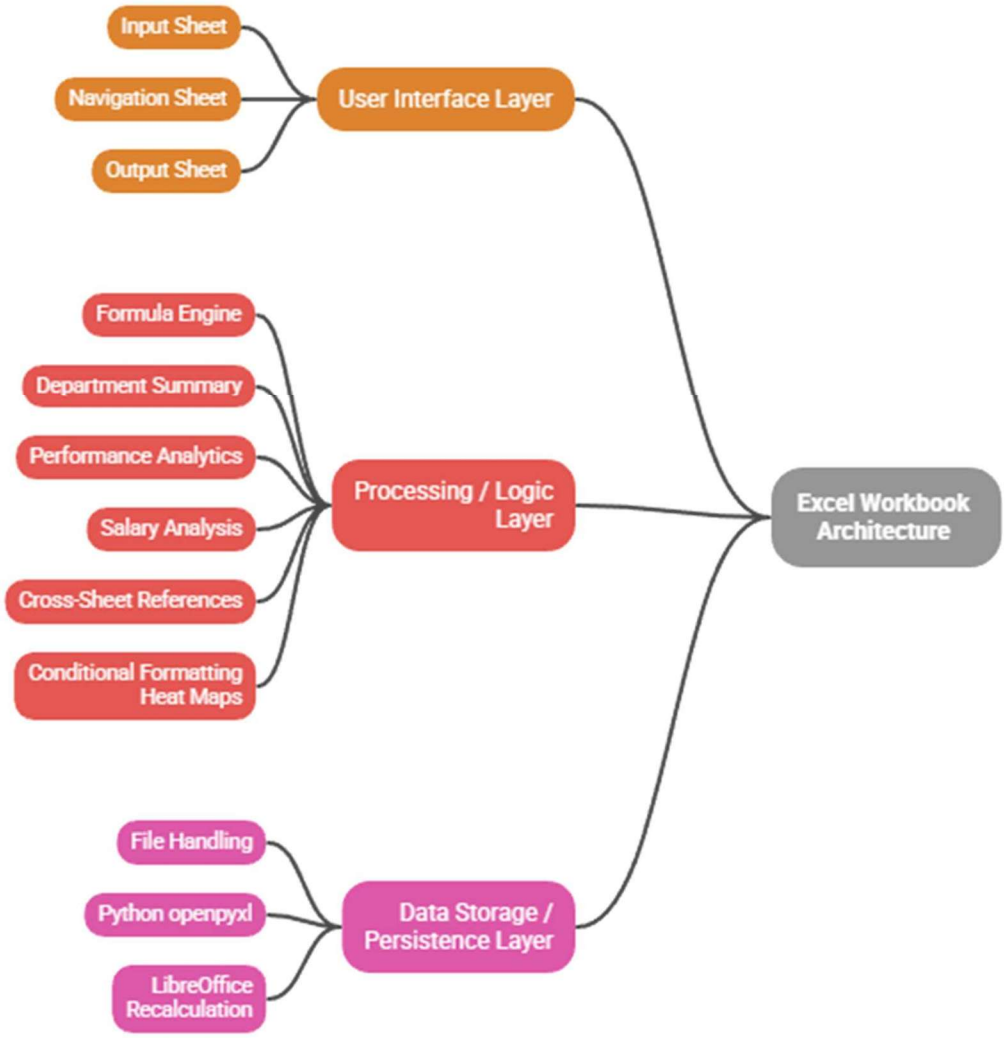
3. Presentation Layer (Salary Analysis & Executive Dashboard):

- Displays **final insights and summaries**.
- Uses functions like **AGGREGATE, AVERAGEIF, STDEV**.
- Provides **salary analysis by level and band**.
- Dashboard shows key metrics like:
  - Total employees
  - Total payroll
  - Average KPI
  - Promotion eligibility
- Designed for **quick decision-making**.

Key Features:

- Automatic data updates across sheets
- Error-free calculations using formulas
- Visual representation using charts and formatting
- Clean and user-friendly design

Excel Workbook Architecture



Functional Modules:

1. Employee Database:

- Main sheet for **data entry and storage**
- Stores employee details (ID, name, salary, performance, etc.)
- Automatically calculates:
  - Bonus % (based on performance)
  - Bonus amount & total compensation
  - Promotion eligibility (using conditions)
  - Salary bands (using IF logic)

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2. Department Summary:

- Generates **department-wise reports**
- Uses functions like **COUNTIF, SUMIF, AVERAGEIF, SUMPRODUCT**
- Displays metrics like **headcount, salary, KPI, attendance**
- Includes **charts for comparison**
- Provides **overall totals and averages**

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3. Performance Analytics:

- Analyzes employee performance data
- Shows:
  - Performance rating distribution
  - Promotion eligibility by level
  - KPI-based performance groups
- Helps in **performance evaluation and decision-making**

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4. Salary Analysis:

- Provides **salary statistics and distribution**
- Includes metrics like **min, max, average, and total cost**
- Groups data by **employment level and salary band**
- Uses functions like **AGGREGATE and AVERAGEIF**

5. Executive Dashboard:

- Displays key insights in **KPI cards format**
- Shows metrics like:
  - Total employees
  - Total payroll
  - Average KPI & attendance
  - Promotion eligibility
- Includes a **quick department summary table**

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6. File Handling / Build Process:

- Workbook created using **Python (openpyxl)**
- Automates **data, formulas, formatting, and charts**
- Uses script to **recalculate and verify formulas**
- Ensures **error-free output**

CHAPTER 4: Implementation

System Components / Tools & Techniques:

1. Platform & Tools:

- Developed using **Microsoft Excel (.xlsx)** and **Python (openpyxl)**
- Uses **LibreOffice** for formula recalculation
- Works on **Windows, Linux, and macOS**

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2. Workbook Structure:

- Contains **5 worksheets** with different tab colors
- Sheets: Employee Database, Department Summary, Performance Analytics, Salary Analysis, Executive Dashboard
- Each sheet has:
  - Proper formatting
  - Frozen headers
  - Adjustable column widths

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3. Core Functions:

- Data entry and storage of employee records
- Automatic calculations (bonus, salary, promotion)
- Department-wise aggregation
- Performance analysis
- Salary benchmarking
- Executive-level reporting

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4. Excel Functions Used:

- **COUNTIF / COUNTIFS:** Count employees based on conditions
- **SUMIF / AVERAGEIF:** Calculate totals and averages
- **SUMPRODUCT:** Multi-condition calculations
- **AGGREGATE:** Error-free min/max calculations
- **IF / Nested IF:** Logic for bonus, salary band, promotion
- **IFERROR:** Handles errors and keeps output clean

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5. Conditional Formatting:

- Applied on **KPI and attendance columns**
- Uses **color scale (Red → Yellow → Green)**
- Helps identify **high and low performers quickly**

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6. Data Visualization:

- Includes **bar and column charts**
- Shows:
  - Department-wise compensation
  - KPI performance comparison
- Improves **data understanding and analysis**

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7. File Handling & Validation:

- Workbook generated using **Python script (openpyxl)**
- Automatically adds:
  - Data, formulas, formatting, charts
- Uses script to **recalculate and verify formulas**
- Ensures **zero errors in all calculation**

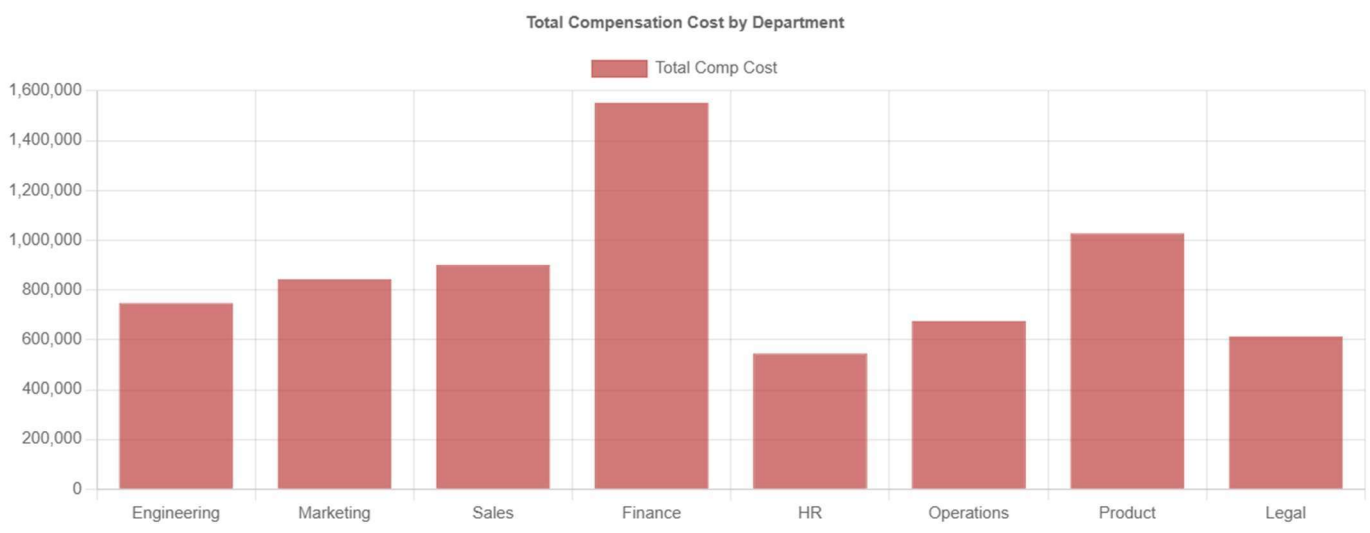
CHAPTER 5: RESULTS / TESTING

The application was tested using multiple input cases to ensure correctness, stability, and data persistence.

Test Case	Description	Expected Result	Outcome
TC1	Add new employee record	Record stored with all 18 fields; formula cells auto-calculate bonus, total comp, eligibility, and band	Passed
TC2	View all employee records	All 50 entries displayed correctly in tabular format with formatting, color coding, and heat maps	Passed
TC3	Department Summary aggregation	COUNTIF, SUMIF, AVERAGEIF values match manual cross-checks for all 8 departments	Passed
TC4	Performance Analytics computation	Rating distribution, level-wise pipeline, and KPI band tables all compute correctly with correct percentages	Passed
TC5	Salary Analysis statistics	Min, Max, Average, StdDev values correct for all 6 levels; AGGREGATE formulas return no errors	Passed
TC6	Executive Dashboard KPI cards	All 8 KPI card values match source data totals and averages; live formulas update when database changes	Passed
TC7	Conditional formatting heat maps	KPI and attendance columns display red-yellow-green gradient correctly for low, mid, and high values	Passed
TC8	Formula error check (all 509 cells)	LibreOffice recalc script returns status: success, total_errors: 0 across all five sheets	Passed

Total Compensation Cost b
'Department Summary&a

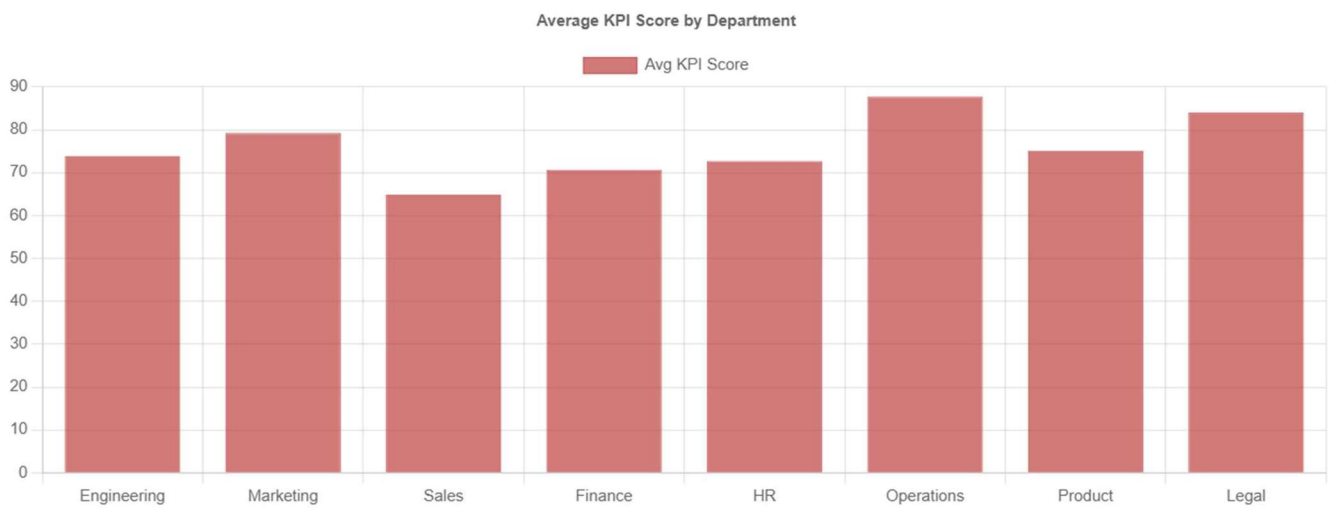
Average KPI Score by Dep
'Department Summary&a



Data Charts 2

Total Compensation Cost b
'Department Summary&a

Average KPI Score by Dep
'Department Summary&a



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EMPLOYEE PERFORMANCE & SALARY MANAGEMENT SYSTEM

Confidential — HR Department | Fiscal Year 2025

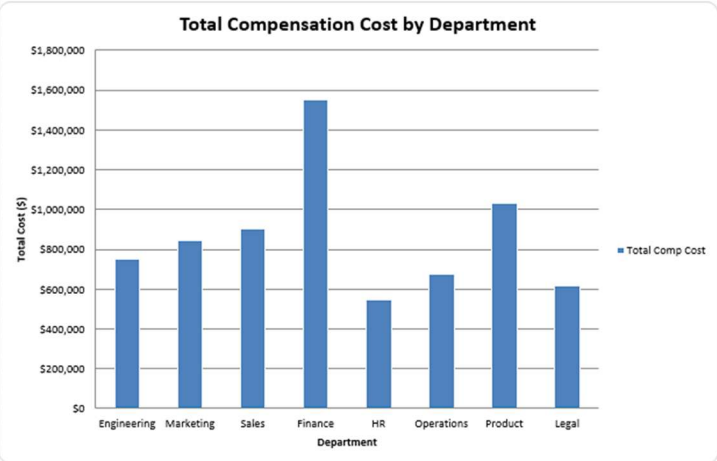
Emp ID	Full Name	Department	Job Title	Level	Hire Date	Years Exp	Base Salary	Bonus %	Bonus Amount	Total Comp	Performance Rating	KPI Score	Attendance %	Training hrs	Promotion Eligible	Salary Band	Status
EMP0001	Jessica Davis	Marketing	SEO Specialist	Junior	2021-11-01	4	$47,800	20.0%	$9,560	$57,360	Outstanding	88.3	90.9%	54	Yes	Band 1 (Junior)	Active
EMP0002	Jessica Thompson	Engineering	DevOps Engineer	Junior	2007-04-01	18	$57,000	2.0%	$1,140	$58,140	Below	87.2	94.0%	28	No	Band 1 (Junior)	Active
EMP0003	Melissa Anderson	HR	Recruiter	Junior	2018-06-01	7	$50,600	8.0%	$4,048	$54,648	Meets	59.6	87.6%	44	No	Band 1 (Junior)	Active
EMP0004	Rebecca Brown	HR	HRBP	Junior	2012-02-01	13	$64,500	20.0%	$12,900	$77,400	Outstanding	79.8	96.6%	46	No	Band 1 (Junior)	Active
EMP0005	Maria Davis	Finance	Financial Analyst	Director	2022-04-01	3	$204,100	2.0%	$4,082	$208,182	Below	94	87.6%	46	No	Band 5 (Director)	Active
EMP0006	William Williams	Operations	Logistics Coordinator	Senior	2019-09-01	6	$114,400	2.0%	$2,288	$116,688	Below	87.8	83.3%	34	No	Band 3 (Senior)	On Leave
EMP0007	Maria Davis	Finance	Controller	Director	2014-07-01	11	$207,500	20.0%	$41,500	$249,000	Outstanding	67	84.2%	40	No	Band 5 (Director)	Active
EMP0008	Michael Wilson	Legal	CLO	Lead	2007-09-01	18	$115,600	20.0%	$23,120	$138,720	Outstanding	66.8	91.7%	51	No	Band 4 (Lead/Mgr)	Active
EMP0009	David Johnson	Finance	FP&A Manager	Mid	2020-11-01	5	$86,500	20.0%	$17,300	$103,800	Outstanding	62.2	93.6%	8	No	Band 2 (Mid)	Active
EMP0010	Charles Wilson	Product	Head of Product	Manager	2024-11-01	1	$168,600	20.0%	$33,720	$202,320	Outstanding	87.4	93.6%	34	No	Band 4 (Lead/Mgr)	Probation
EMP0011	Melissa Garcia	Operations	Process Manager	Junior	2016-09-01	9	$54,100	15.0%	$8,115	$62,215	Exceeds	89.3	90.2%	80	Yes	Band 1 (Junior)	Active
EMP0012	Emily Lee	Finance	Controller	Mid	2008-01-01	17	$88,800	15.0%	$13,320	$102,120	Exceeds	82	89.8%	46	Yes	Band 2 (Mid)	Probation
EMP0013	Jessica Moore	HR	HR Generalist	Mid	2022-12-01	3	$88,700	20.0%	$17,740	$106,440	Outstanding	76.9	81.4%	16	No	Band 2 (Mid)	Active
EMP0014	William Thompson	Legal	Compliance Officer	Manager	2011-04-01	14	$163,600	8.0%	$13,088	$176,688	Meets	96.8	95.1%	39	Yes	Band 4 (Lead/Mgr)	Active
EMP0015	Sarah Davis	Operations	COO	Lead	2014-01-01	11	$119,000	8.0%	$9,520	$128,520	Meets	81.5	84.6%	0	Yes	Band 4 (Lead/Mgr)	Active
EMP0016	Maria Anderson	Engineering	Software Engineer	Mid	2017-05-01	8	$66,800	8.0%	$5,344	$72,144	Meets	85.1	84.3%	73	Yes	Band 2 (Mid)	Active
EMP0017	Melissa Miller	Legal	CLO	Mid	2011-06-01	14	$67,400	15.0%	$10,110	$77,510	Exceeds	74.1	89.3%	12	No	Band 2 (Mid)	Active
EMP0018	Sarah Davis	Product	Product Analyst	Director	2007-08-01	18	$163,400	2.0%	$3,268	$166,668	Below	61.3	83.7%	31	No	Band 4 (Lead/Mgr)	Probation
EMP0019	Maria Lee	Marketing	Marketing Analyst	Lead	2022-04-01	3	$143,400	8.0%	$11,472	$154,872	Meets	62.5	89.7%	51	No	Band 4 (Lead/Mgr)	Active
EMP0020	Richard Wilson	Sales	Account Executive	Lead	2010-05-01	15	$147,100	20.0%	$29,420	$176,520	Outstanding	74	99.1%	62	No	Band 4 (Lead/Mgr)	Active
EMP0021	Maria Moore	Finance	Accountant	Senior	2023-12-01	2	$119,400	2.0%	$2,388	$121,788	Below	69.1	81.0%	64	No	Band 3 (Senior)	Probation
EMP0022	Emily Williams	Sales	Account Executive	Junior	2017-07-01	8	$56,900	8.0%	$4,552	$61,452	Meets	60.4	97.8%	74	No	Band 1 (Junior)	Active
EMP0023	Melissa Moore	Engineering	Software Engineer	Manager	2014-05-01	11	$148,300	8.0%	$11,864	$160,164	Meets	64.2	94.3%	33	No	Band 4 (Lead/Mgr)	Active
EMP0024	Thomas Anderson	Sales	Business Dev Manager	Director	2022-01-01	3	$215,000	15.0%	$32,250	$247,250	Exceeds	75.6	100.0%	9	No	Band 5 (Director)	Active
EMP0025	Michael Thomas	Finance	Controller	Manager	2017-06-01	8	$164,000	20.0%	$32,800	$196,800	Outstanding	67.8	88.8%	38	No	Band 4 (Lead/Mgr)	Active

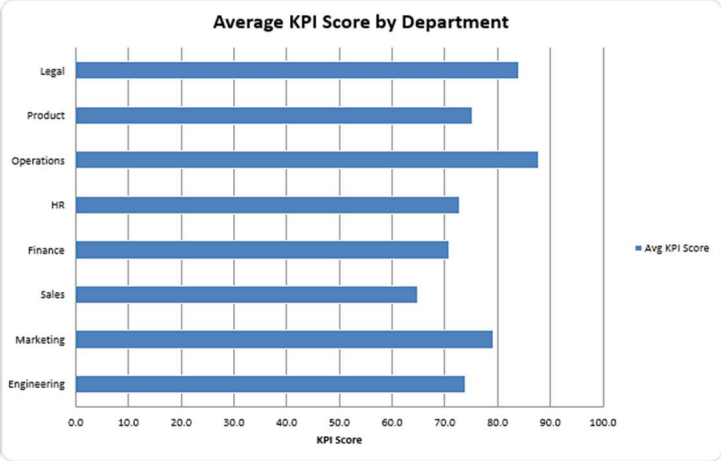
DEPARTMENT PERFORMANCE SUMMARY

Auto-calculated from Employee Database | Use slicers/filters for drill-down

Department	Head Count	Avg Base Salary	Total Payroll	Avg Bonus %	Total Bonus	Total Comp Cost	Avg KPI Score	Avg Attendance %	Eligible for Promo
Engineering	6	$112,900	$677,400	9.0%	$70,992	$748,392	73.9	89.1%	1
Marketing	6	$130,250	$781,500	9.2%	$62,462	$843,962	79.3	90.0%	1
Sales	5	$160,820	$804,100	11.8%	$97,030	$901,130	64.9	95.5%	0
Finance	12	$114,825	$1,377,900	13.1%	$174,738	$1,552,638	70.7	90.3%	1
HR	5	$95,580	$477,900	14.0%	$67,746	$545,646	72.7	86.4%	1
Operations	6	$104,050	$624,300	9.3%	$51,858	$676,158	87.7	86.7%	5
Product	6	$153,267	$919,600	12.0%	$108,602	$1,028,202	75.2	90.5%	1
Legal	4	$134,750	$539,000	14.5%	$75,178	$614,178	84.0	89.6%	2
TOTAL / AVERAGE	50	$125,805	$6,201,700	11.6%	$708,606	$6,910,306	76.0	89.7%	12

Total Compensation Cost by Department





Employee Database | Department Summary | Performance Analytics | Executive Dashboard | Salary A

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A1 PERFORMANCE ANALYTICS DASHBOARD

PERFORMANCE ANALYTICS DASHBOARD

Performance Rating Distribution			
Rating	Count	% of Total	Avg KPI
Outstanding	16	32.0%	76.8
Exceeds	7	14.0%	80.7
Meets	17	34.0%	71.5
Below	10	20.0%	76.1
Total	**50**	**100.0%**	**75.4**

Promotion Eligibility Pipeline				
Level	Total	Eligible	% Eligible	Avg Salary
Junior	8	2	25.0%	$55,650
Mid	8	3	37.5%	$79,038
Senior	6	0	0.0%	$112,867
Lead	10	3	30.0%	$131,950
Manager	8	3	37.5%	$153,888
Director	10	1	10.0%	$189,640

KPI Threshold Breakdown			
KPI Range	Employee Count	% Share	Avg Total Comp
90-100 (Top)	7	14.0%	$178,748
80-89 (Good)	12	24.0%	$115,695
70-79 (Average)	9	18.0%	$130,668
Below 70 (At Risk)	22	44.0%	$140,669

A1 EXECUTIVE HR DASHBOARD

EXECUTIVE HR DASHBOARD

Real-time KPIs — Linked from source sheets | FY2025

Total Employees	Avg KPI Score	Total Payroll	Total Comp Cost
50	75.4	$6,201,700	$6,910,306

Promo Eligible	Avg Attendance %	Outstanding Perf	Avg Total Comp
12	89.8%	16	$138,206

Department Quick View					
Department	Head Count	Avg KPI	Avg Attendance	Total Comp	Promo Count
Engineering	6	73.9	89.1%	$748,392	1
Marketing	6	79.3	90.0%	$843,962	1
Sales	5	64.9	95.5%	$901,130	0
Finance	12	70.7	90.3%	$1,552,638	1
HR	5	72.7	86.4%	$545,646	1
Operations	6	87.7	86.7%	$676,158	5
Product	6	75.2	90.5%	$1,028,202	1
Legal	4	84.0	89.6%	$614,178	2

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A1 | COMPENSATION & SALARY ANALYSIS

COMPENSATION & SALARY ANALYSIS

Salary Statistics by Level

Level	Count	Min Salary	Max Salary	Avg Salary	Median (Approx)	Std Dev	Total Cost
Junior	8	$47,800	$64,500	$55,650	$56,400	$5,055	$497,863
Mid	8	$66,800	$89,600	$79,038	$79,850	$10,247	$739,369
Senior	6	$91,500	$124,900	$112,867	$116,900	$12,426	$711,858
Lead	10	$115,400	$147,100	$131,950	$137,300	$12,688	$1,482,486
Manager	8	$122,100	$168,800	$153,888	$158,400	$16,068	$1,379,334
Director	10	$163,400	$215,000	$189,640	$194,400	$18,456	$2,099,396
Company Total	**50**	**$47,800**	**$215,000**	**$124,034**	**$131,400**	**$48,057**	**$6,910,306**

Salary Band Distribution

Band	Range	Employee Count	% of Workforce	Avg KPI	Avg Attendance	Avg Total Comp	Promo Eligible
Band 1 (Junior)	<$65K	8	16.0%	74.8	92.6%	$62,233	2
Band 2 (Mid)	$65K-$90K	8	16.0%	76.3	90.4%	$92,421	3
Band 3 (Senior)	$90K-$130K	6	12.0%	70.6	89.4%	$118,643	0
Band 4 (Lead/Mgr)	$130K-$170K	21	42.0%	75.8	88.8%	$162,823	6
Band 5 (Director)	$170K+	7	14.0%	77.9	89.2%	$220,275	1

CHAPTER 6: Conclusion

Conclusion:

- The system successfully provides a **simple and reliable HR data management solution**.
- Uses **five interconnected Excel sheets** for complete data handling and analysis.
- Ensures **automatic updates** using formula-based calculations.
- Eliminates errors by avoiding **manual data handling**.
- Demonstrates effective use of **advanced Excel functions** (COUNTIF, SUMIF, etc.).
- Integrates **Python (openpyxl)** for automated workbook creation.
- Ensures **accuracy and reliability** through formula validation.
- Includes **charts and dashboards** for better visualization.
- Acts as a **complete offline HR analytics tool** for small to medium organizations.

Future Scope:

- Develop a **web-based application** (React/Vue) for online access
- Integrate with **databases** (MySQL, PostgreSQL) for large data handling
- Connect with **Power BI / Tableau** for advanced dashboards
- Add **multi-user login and role-based access**
- Enable **report export (PDF/Excel)**
- Use **machine learning** for prediction (attrition, salary trends)

CHAPTER 7: References

- E. Balagurusamy, “Object Oriented Programming with C++”, Tata McGraw-Hill Education. (Referenced for foundational software design principles and modular architecture concepts applied in workbook layer design.)
- Bjarne Stroustrup, “The C++ Programming Language”, Addison-Wesley Professional. (Referenced for systematic design methodology and separation-of-concerns principles.)
- Microsoft Corporation, “Excel Documentation and Function Reference”, Microsoft Learn, 2024. Available at: https://learn.microsoft.com/en-us/office/client-developer/excel — Primary reference for all Excel function syntax, conditional formatting APIs, and charting APIs used in this project.
- openpyxl Development Team, “openpyxl Library Documentation — A Python Library to Read/Write Excel 2010 xlsx/xlsm/xltx/xltm Files”, Version 3.1+, 2024. Available at: https://openpyxl.readthedocs.io — Used as the primary programming interface for workbook generation.
- Walkenbach, J., “Excel 2019 Bible”, John Wiley & Sons, Inc. — Referenced for advanced formula construction techniques, AGGREGATE function usage, conditional formatting best practices, and embedded chart design patterns.
- www.exceljet.net — Excel Formula Reference, Function Examples, and Best Practices. Used for verifying SUMPRODUCT multi-condition counting patterns and AVERAGEIFS multi-criteria formula structure.
- www.chandoo.org — Excel Dashboard Design and Conditional Formatting Tutorials. Referenced for KPI card dashboard layout principles and color-scale conditional formatting implementation techniques.
- www.geeksforgeeks.org — Python openpyxl Tutorials, PatternFill and Font Styling Examples, and BarChart Configuration References. Used during implementation of formatting and charting modules.
- Stack Overflow Community Discussions on Cross-Sheet SUMIF References, AGGREGATE Function Error Handling, openpyxl Chart Series Configuration, and LibreOffice Headless Recalculation. Available at: https://stackoverflow.com